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Section 1: Technical Understanding

The most surprising discovery for me was understanding how different types of operations affect images in completely different ways. Initially, I assumed that most image edits worked the same way, but learning the difference between point operations and neighborhood operations changed that. Point operations work on individual pixels one at a time, which explains why brightness and contrast changes feel global and uniform. However, neighborhood operations consider surrounding pixels, which is why blurring and edge detection can detect structure and shape instead of just changing intensity. The mathematical operations that were used directly relate to the visual effects we see. Simple math like adding or scaling pixel values creates brightness and contrast changes, while convolution uses matrices to blend neighboring pixel values together. Seeing these equations applied visually made it clear that image processing is really controlled math producing predictable results. The most challenging technique to understand was convolution used in neighborhood operations. At first, it was confusing how a small kernel could influence the entire image, but seeing how it slides across the image and combines pixel values helped it click. Once I understood that each output pixel is a weighted combination of its neighbors, edge detection and blurring made a lot more sense.

Section 2: Connections and Applications

Although we did not have an in-class Nano Banana demonstration, I looked it up to understand how it works. It became clear that Nano Banana relies on the same image processing techniques used in this lab. Operations like contrast enhancement, filtering, and edge detection form the foundation that AI systems build on to automatically analyze and modify images. The techniques from this lab have many real-world applications. They can be used in medical imaging to enhance scans (which is what I'm interested in), in security systems to improve video clarity, in photography to adjust lighting and detail, and in machine learning pipelines to prepare image data for training. In a future project, I could combine traditional image processing with AI by using filters and contrast adjustments to clean and enhance images before feeding them into a machine learning model. This would improve the quality of the data while allowing AI to handle higher-level analysis and pattern recognition.

Section 3: Personal Reflection

The area of image processing that interests me most for further exploration is medical imaging. I would like to learn how techniques like contrast enhancement and edge detection are used to make important features in X-rays, MRIs, or CT scans easier to see. This lab changed how I understand digital photography and image editing. Instead of seeing edits as artistic adjustments, I now see them as mathematical operations applied to pixel data. That understanding makes image editing feel more logical and controlled. I still have questions about how these traditional techniques scale when used with deep learning models and how AI

systems decide which operations to apply automatically. Exploring how classical image processing and AI complement each other is something I would like to learn more about.